

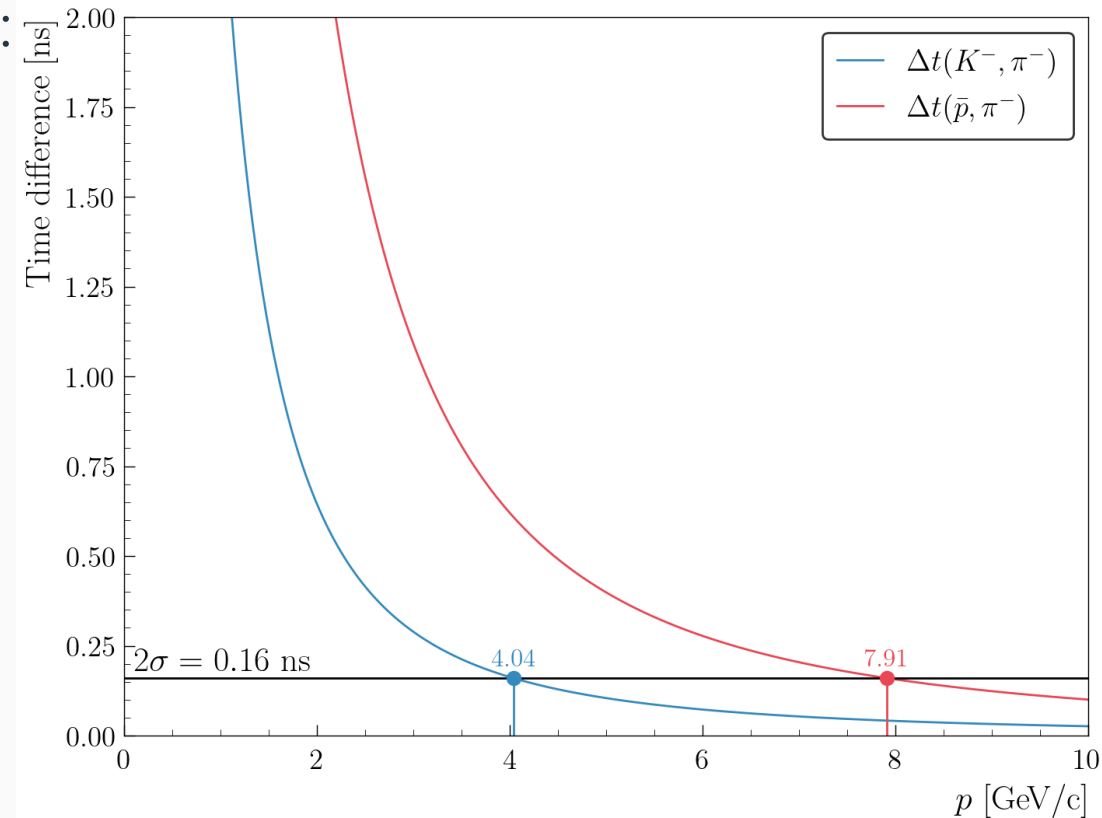
Hadron PID with AI

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- The hadron identification is done after the identification of the trigger electron and use the start time define by the trigger electron.
- It use mostly the FTOF and rely on β (or Δt):
- FTOF can separate kaon and pion up to 4 GeV with a 2σ resolution of the FTOF (5.7 GeV for 1σ), and proton and pion up to 8 GeV.
- Above that, the FTOF is not enough
- We can try to use other detectors:
 - Such as calorimeters and Cherenkov counters



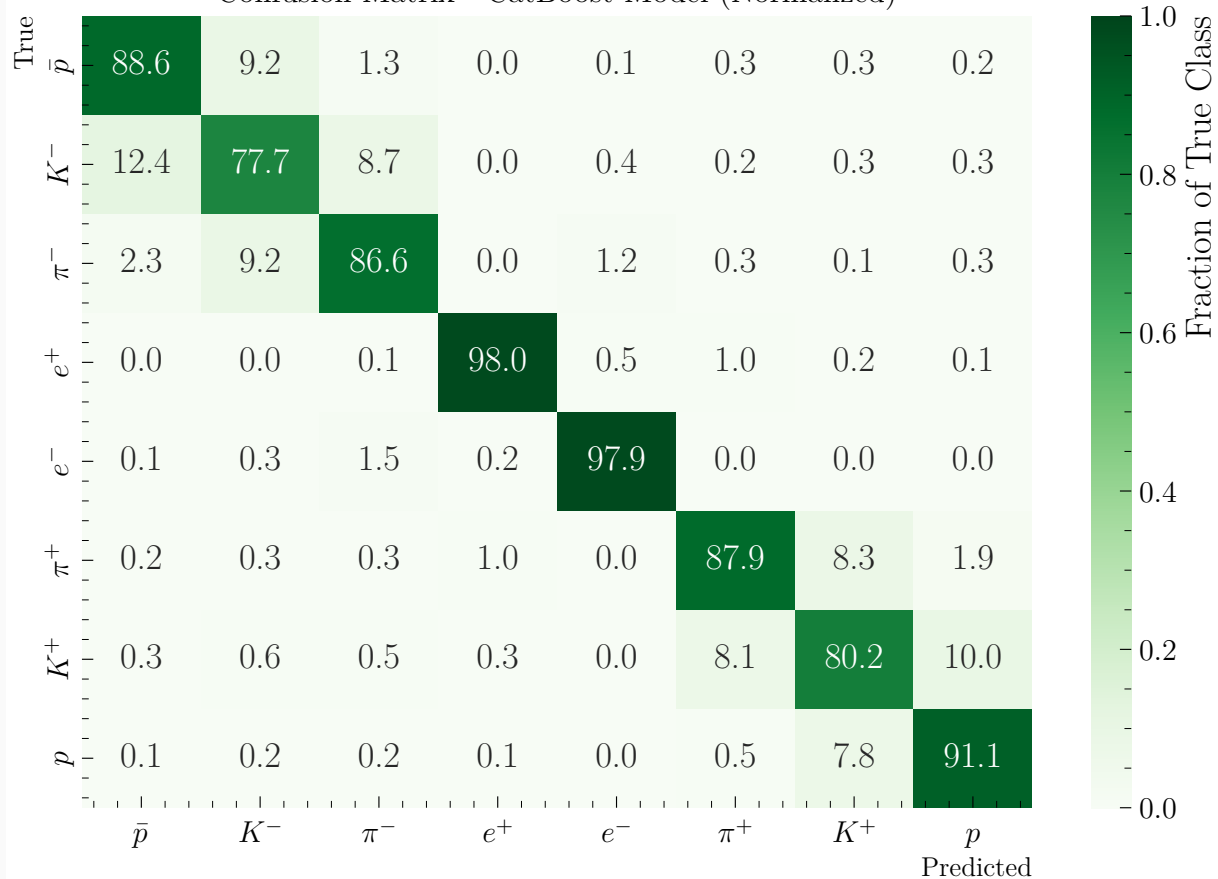
- To use the information of all these detectors in the most efficient way, we can use AI. More specifically the Boosted decision tree architecture are well suited for that problem.
- The goal is to have a model that use information only from the REC banks.
- So, it can be use in the analysis step (on DST files).
- Use information from the calorimeters, Cherenkov counters, TOF and higher-level variables (px, py, pz, charge, ...)
- Use all variable found in the DST banks: 96 features.

- Try many different architectures: Gradient Boosting Decision Trees (GBDT), XGBoost, LightGBM and CatBoost (same family), MLP, MLP + GBDT.
- The best results are for CatBoost (Categorical Boosting). Principle: Builds an ensemble of "weak" models (trees) to describe complex non-linear systems. Goal: Mapping observables (Detector readings, momentum) → Properties (Particle ID).
- Standard method could fail when data is noisy or too high-dimensional. CatBoost learns the 'shape' of the physics directly from the experimental data (MC).
- Uses Symmetric Trees (balanced structure). This makes inference very fast.

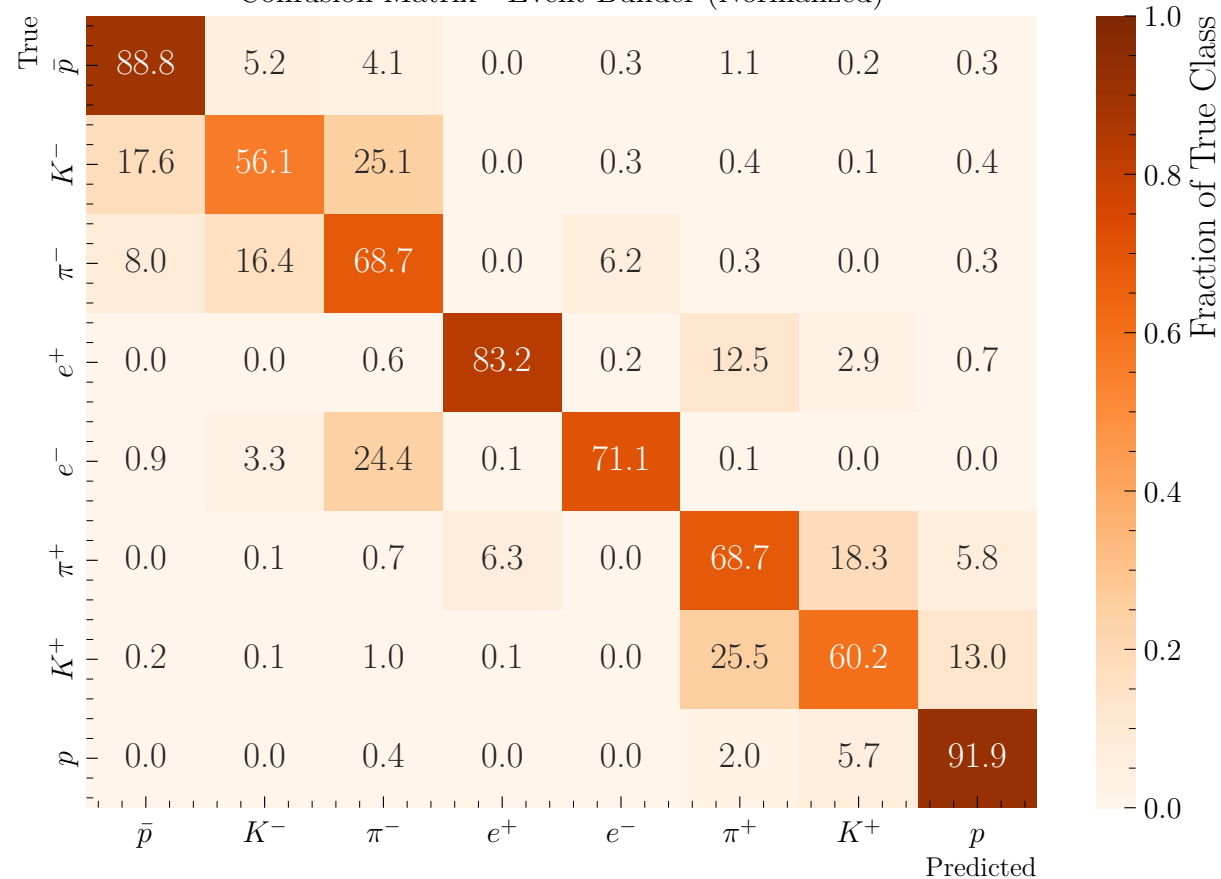


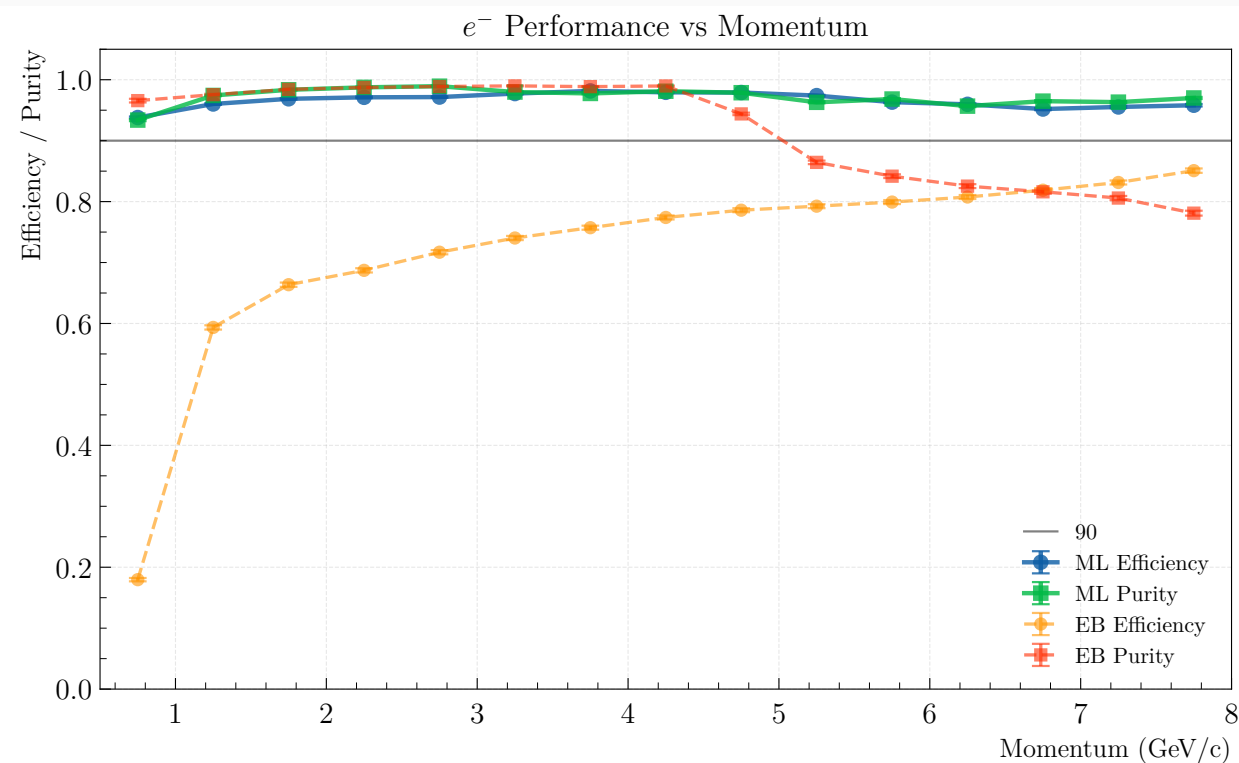
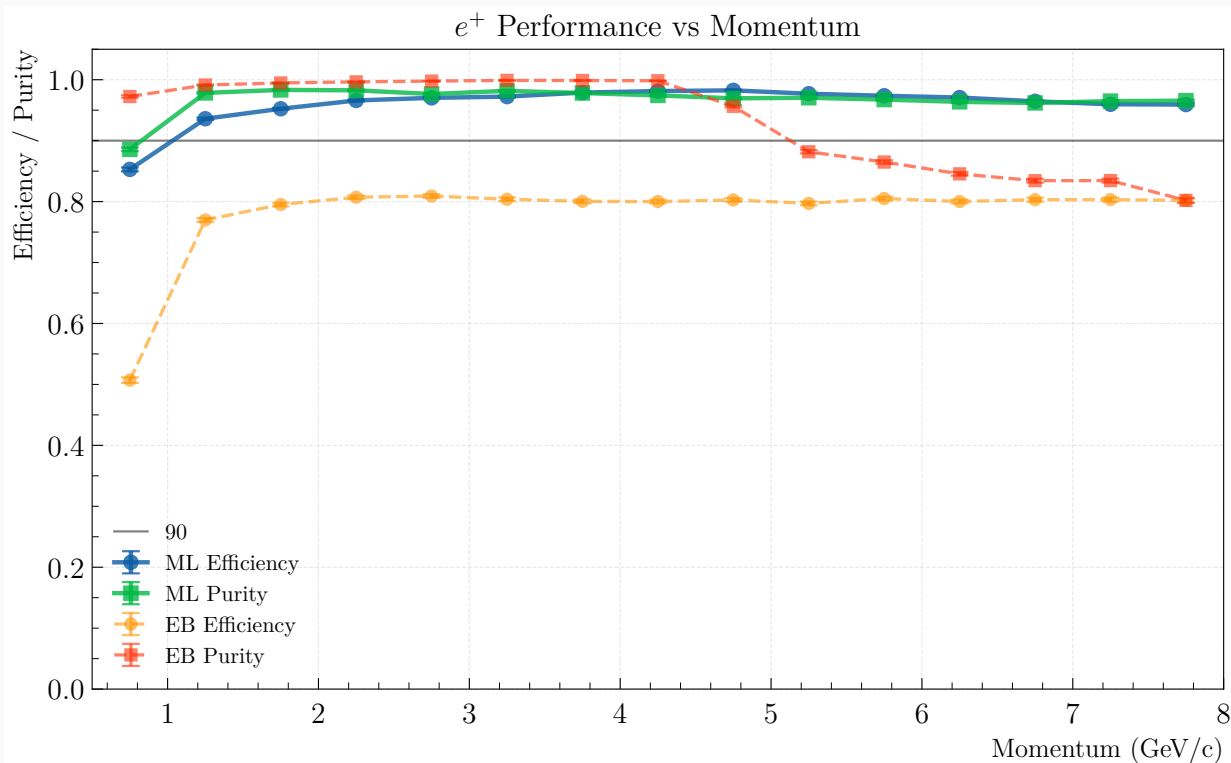
Hadron identification

Confusion Matrix - CatBoost Model (Normalized)



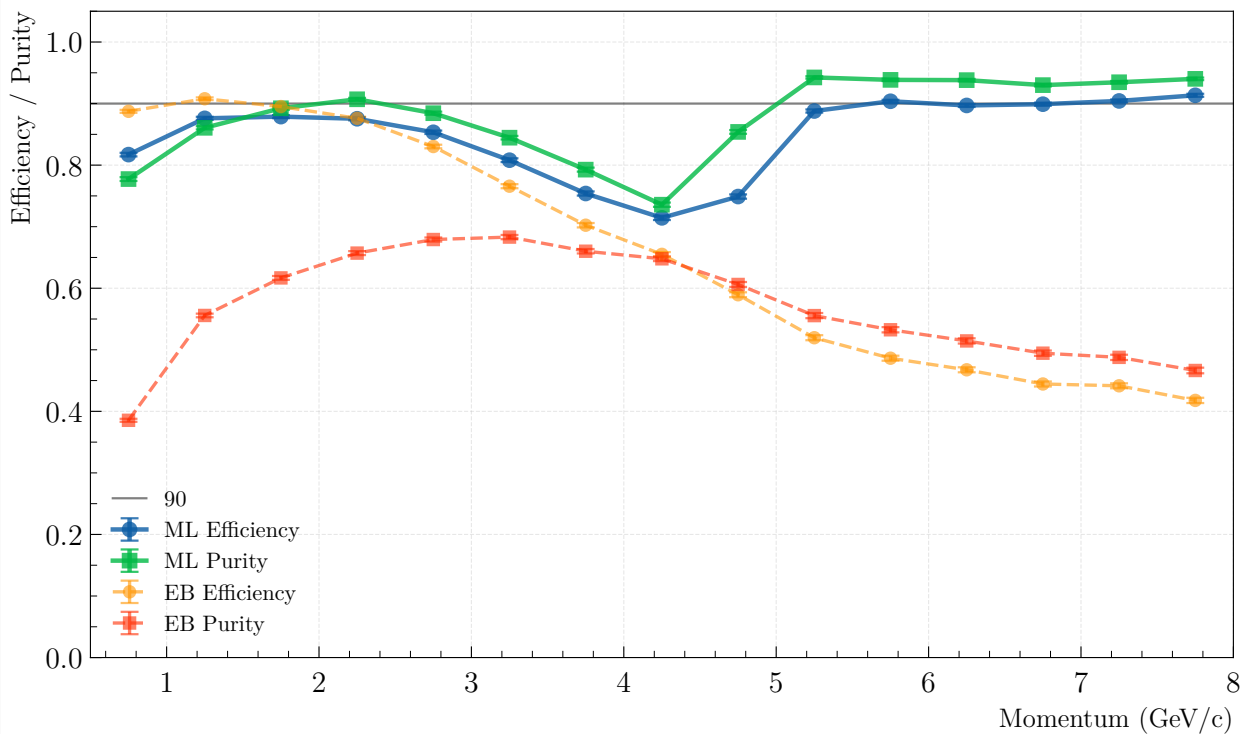
Confusion Matrix - Event Builder (Normalized)



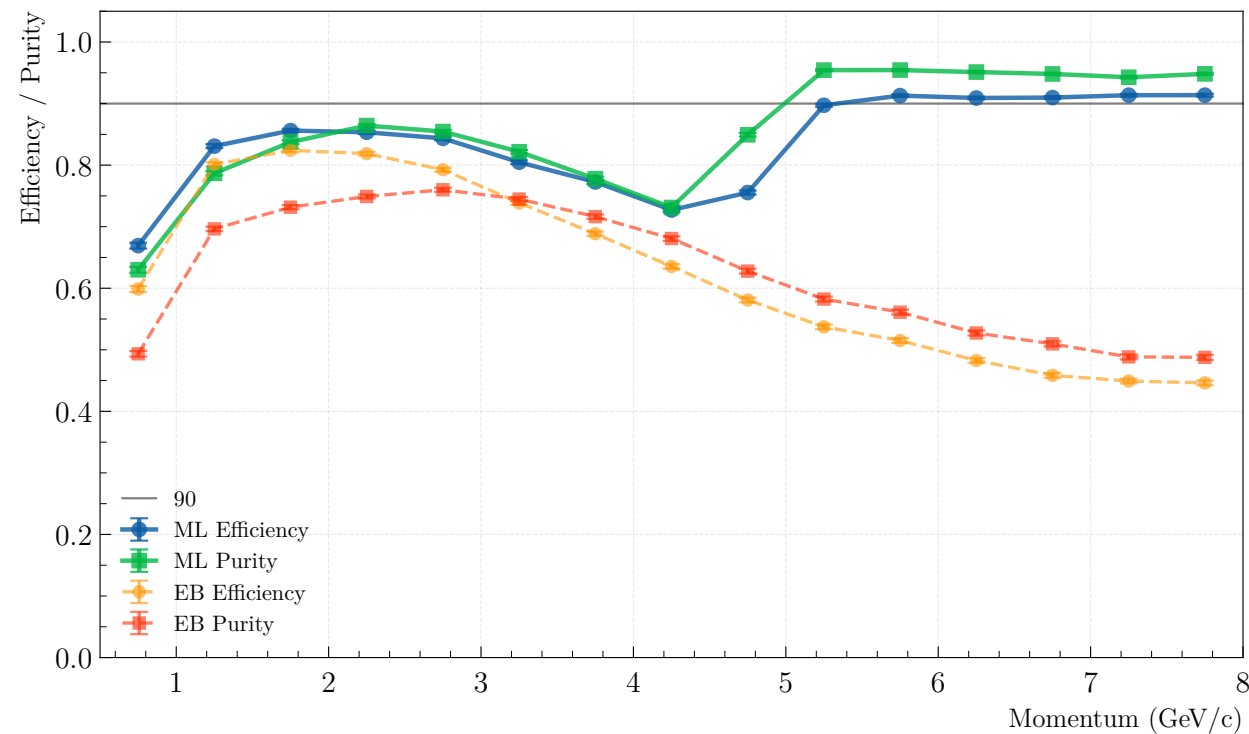


- Electron here are any electron that is not the trigger

π^- Performance vs Momentum



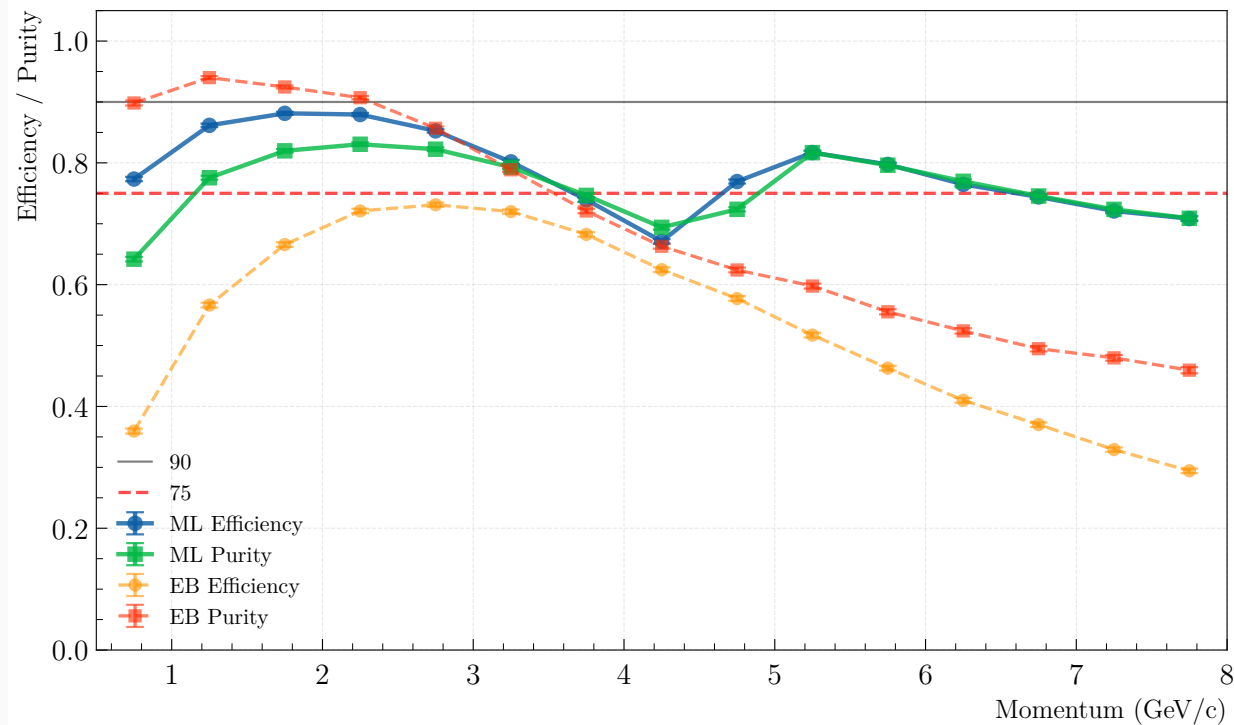
π^+ Performance vs Momentum



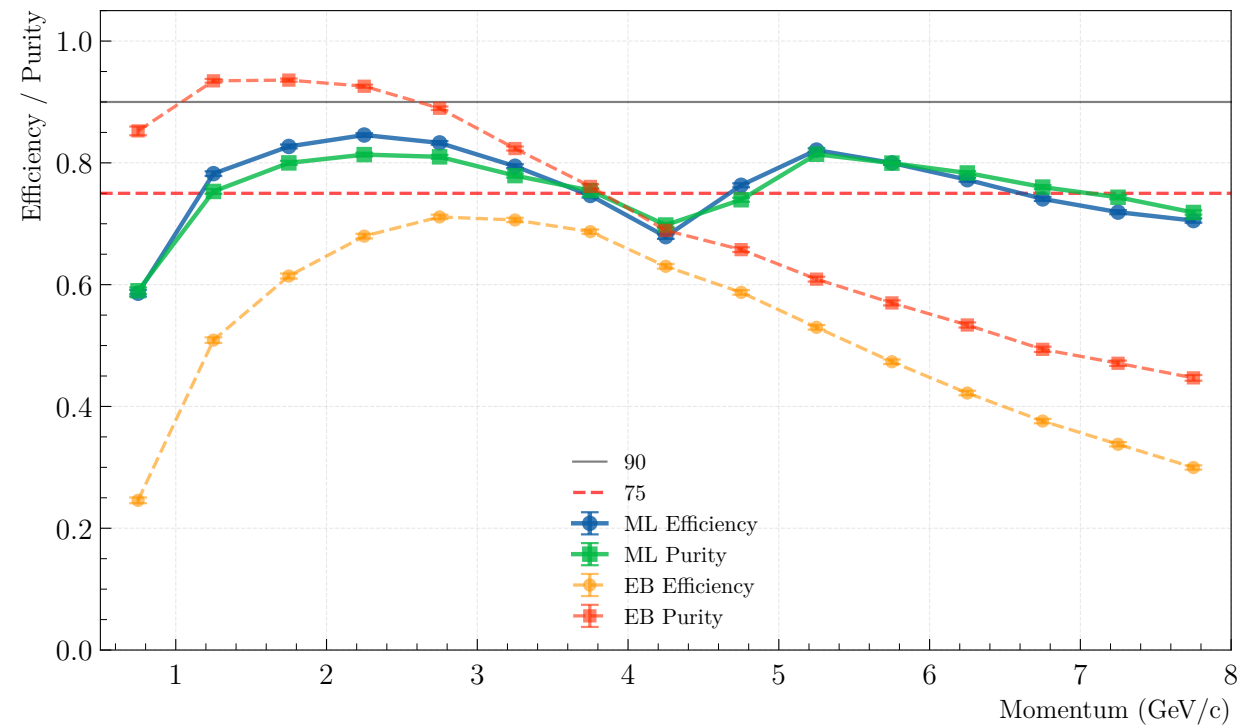


Hadron identification

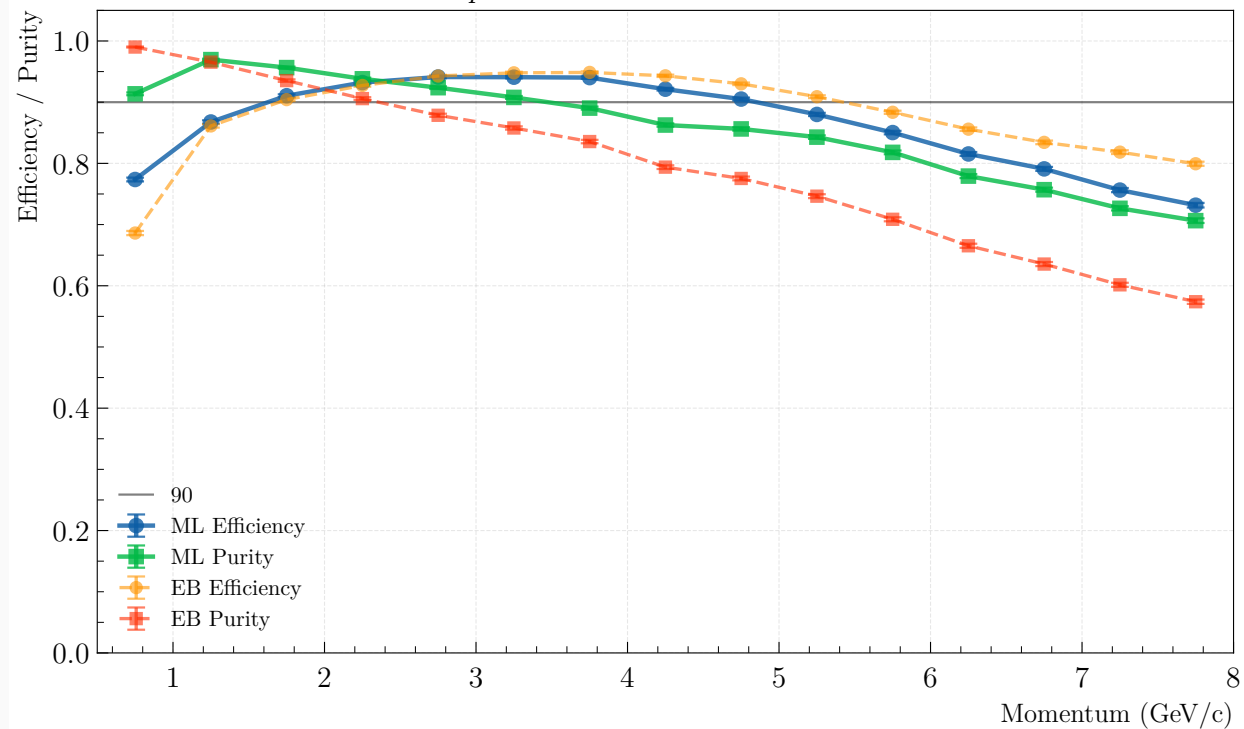
K^- Performance vs Momentum



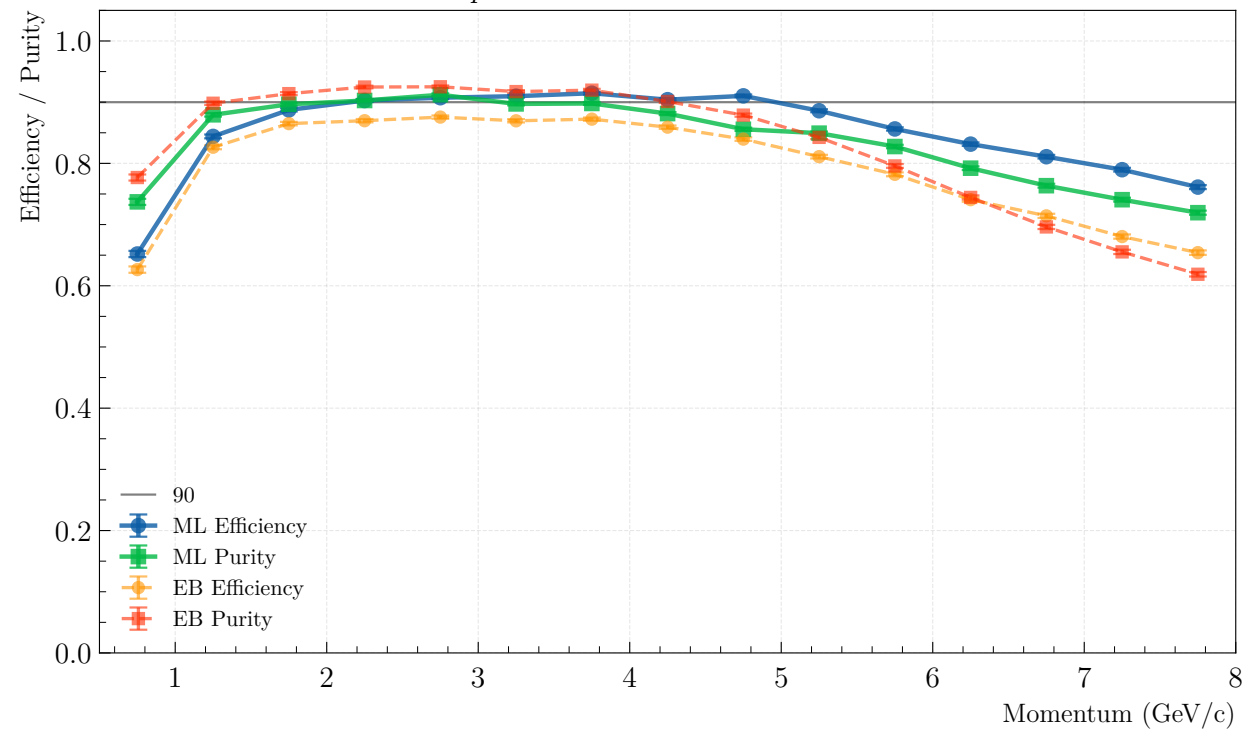
K^+ Performance vs Momentum



$\bar{p}$ Performance vs Momentum



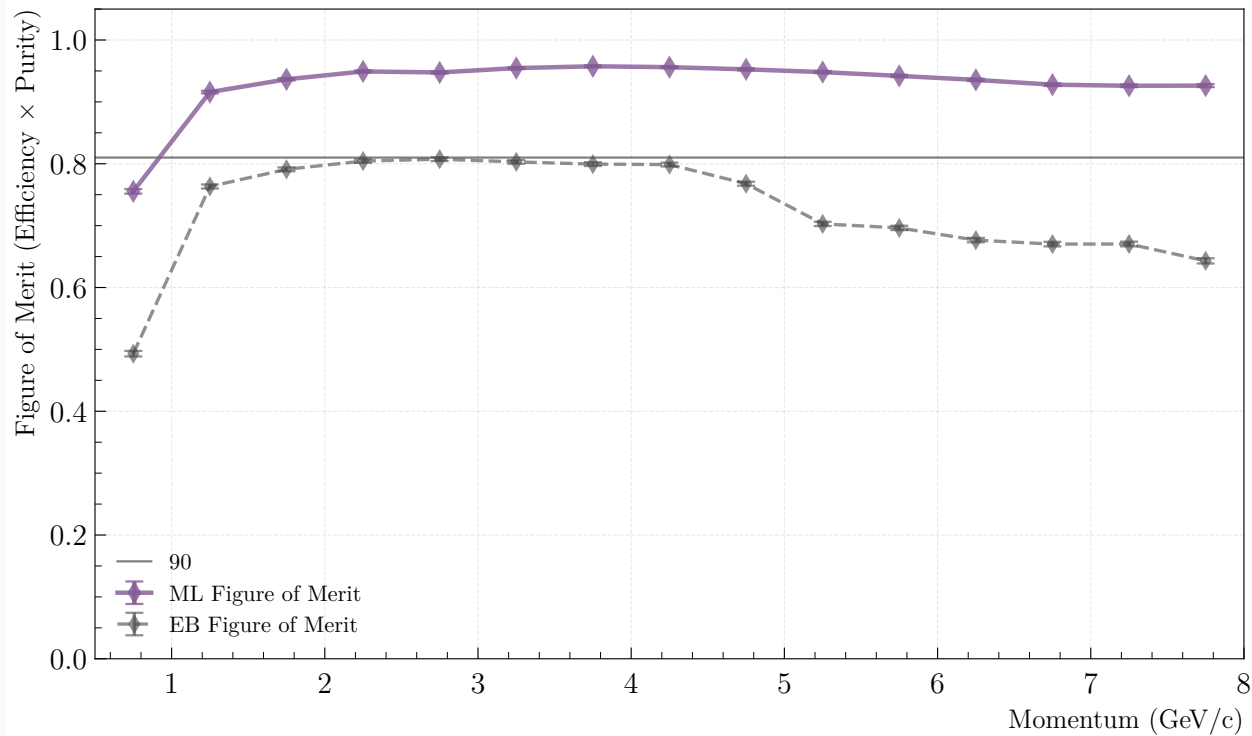
p Performance vs Momentum



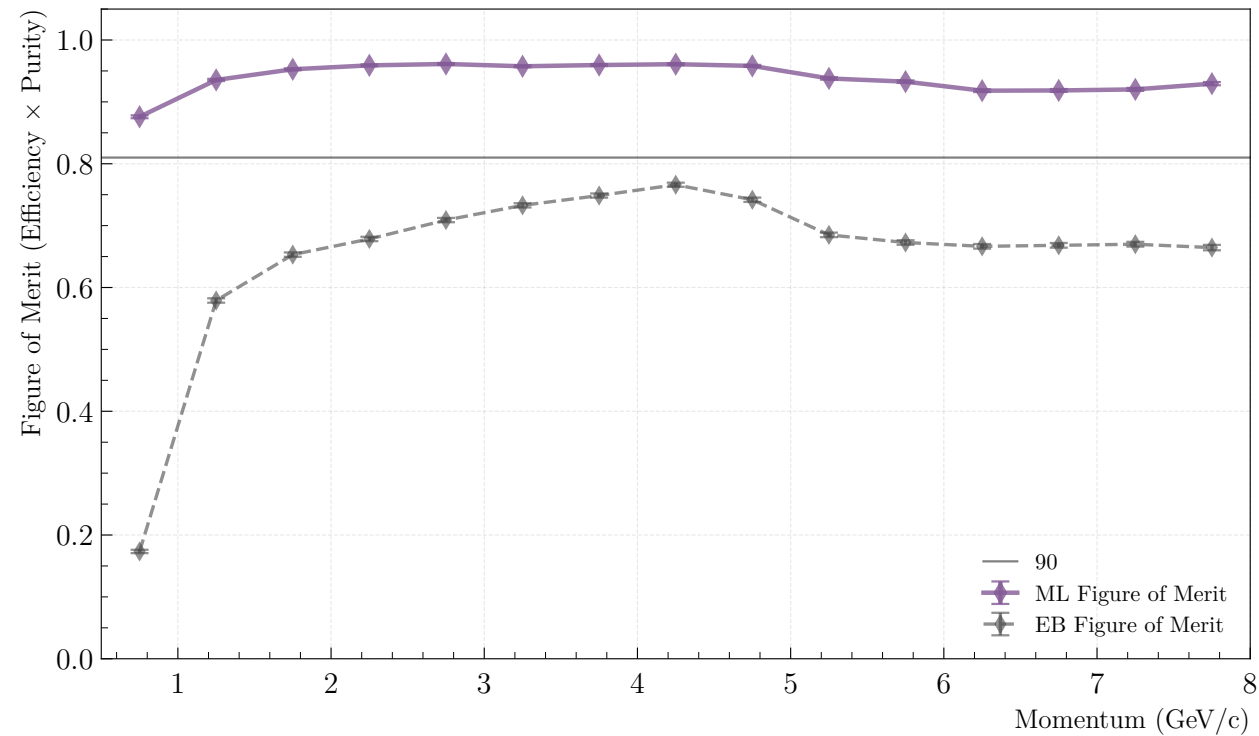


Hadron identification

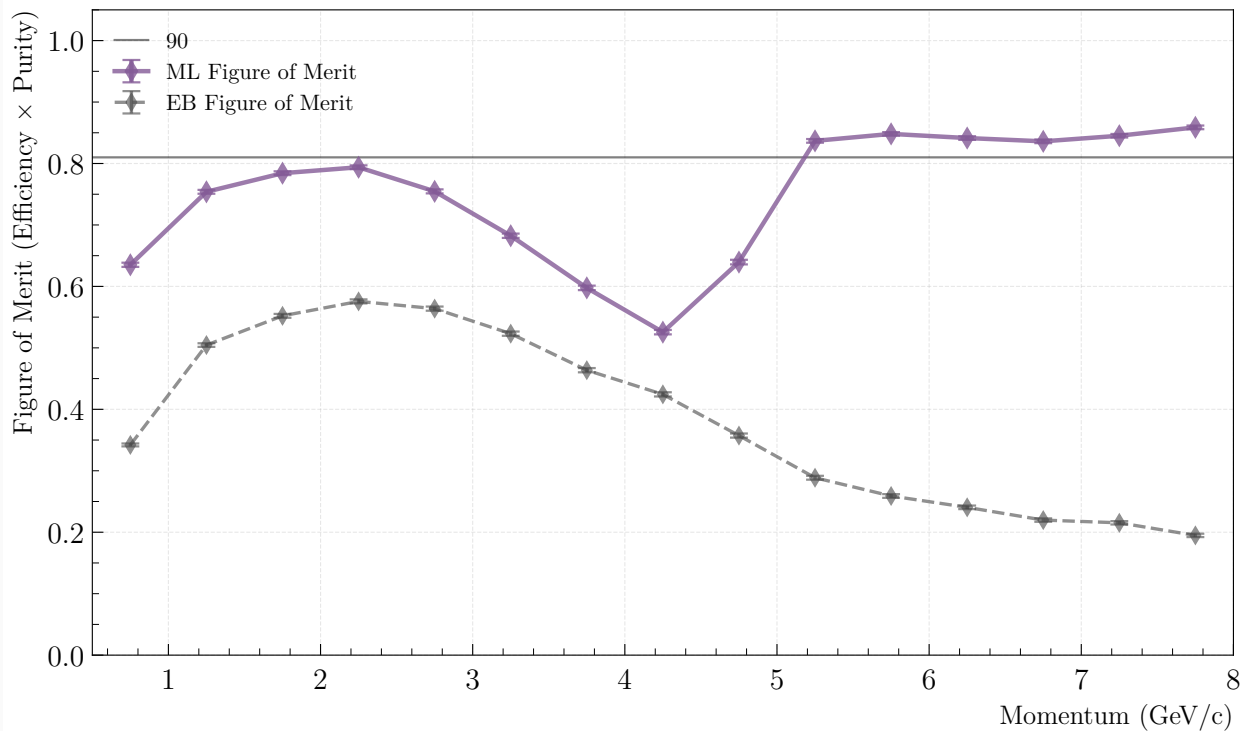
e^+ FOM vs Momentum



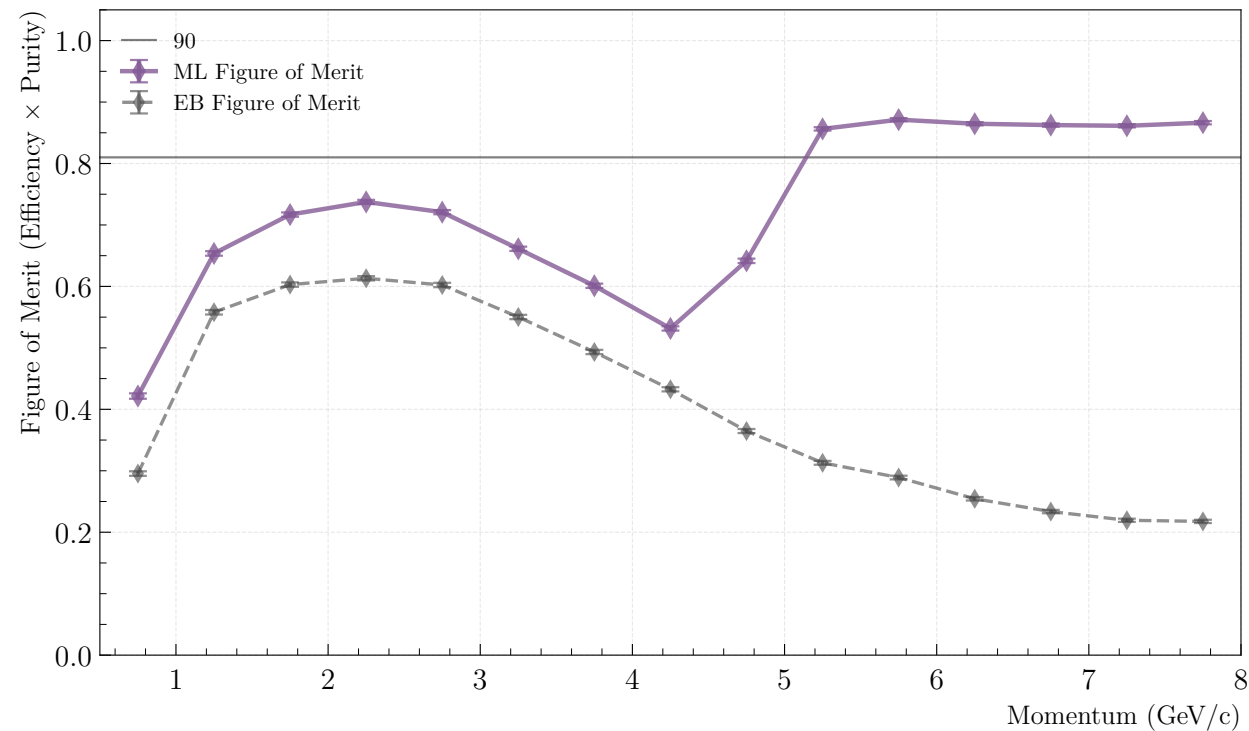
e^- FOM vs Momentum



π^- FOM vs Momentum



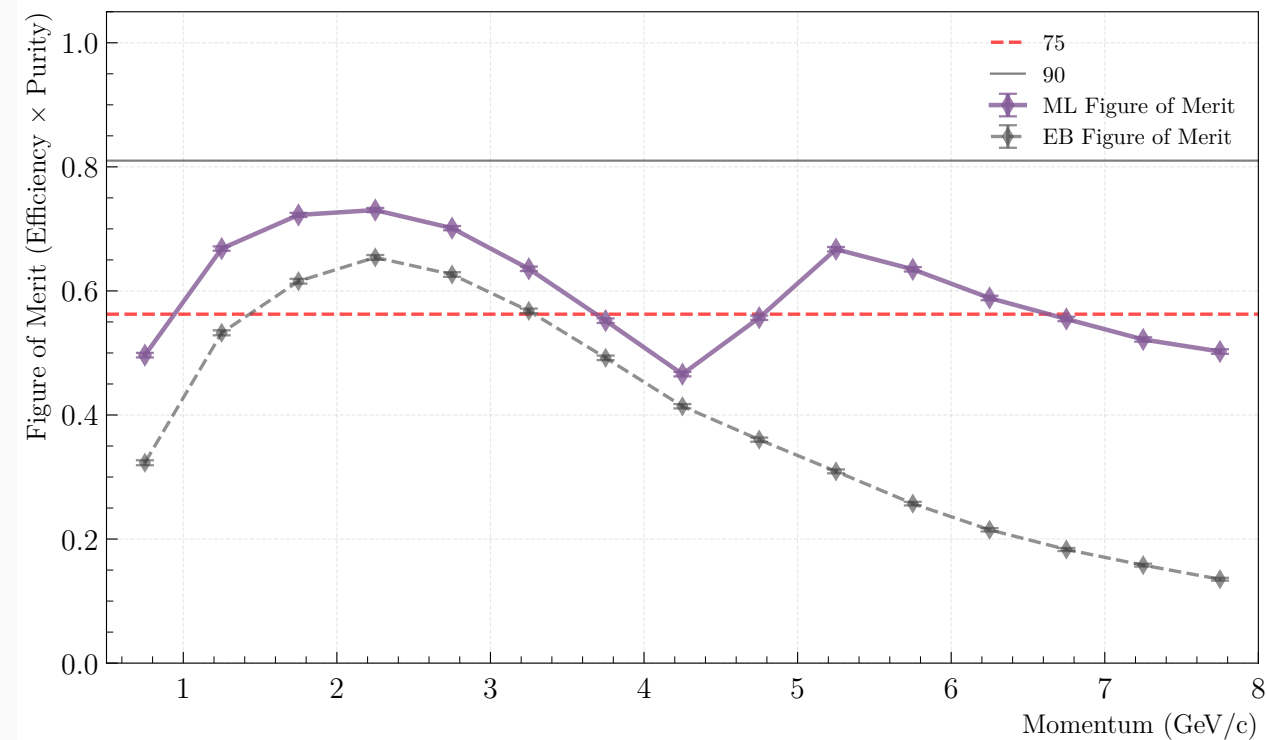
π^+ FOM vs Momentum



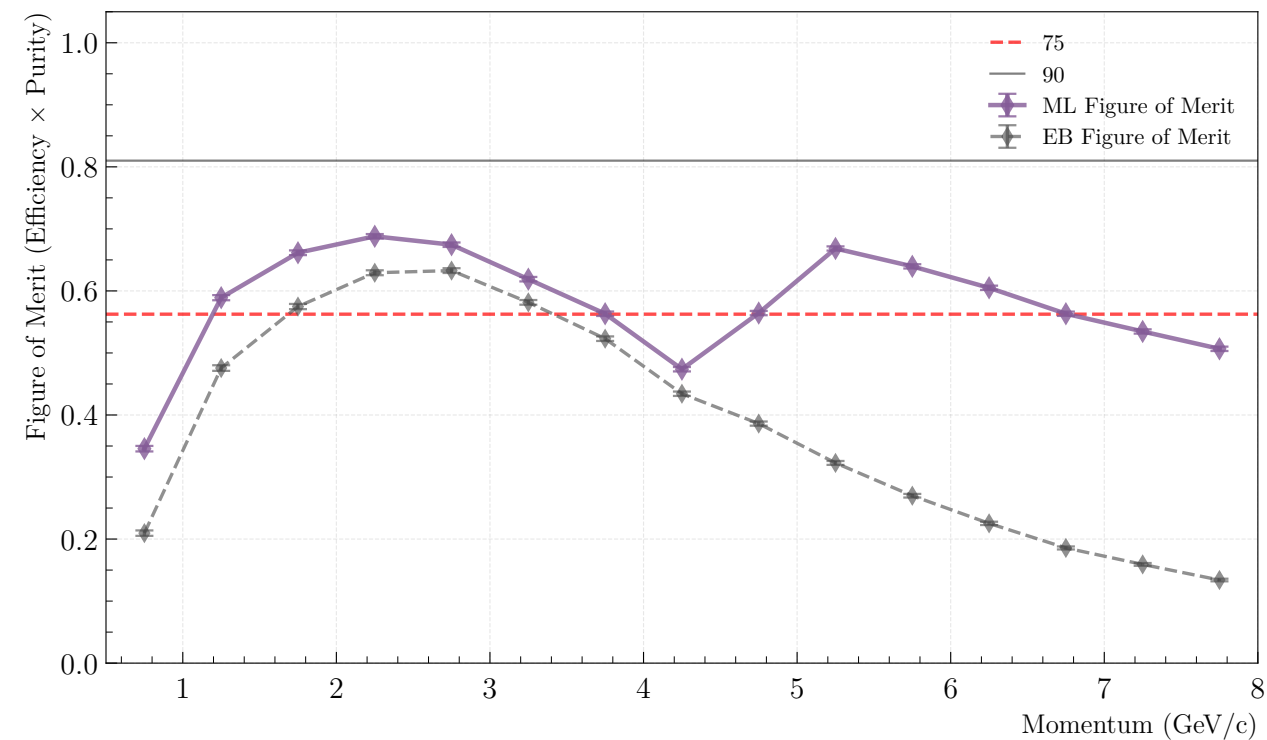


Hadron identification

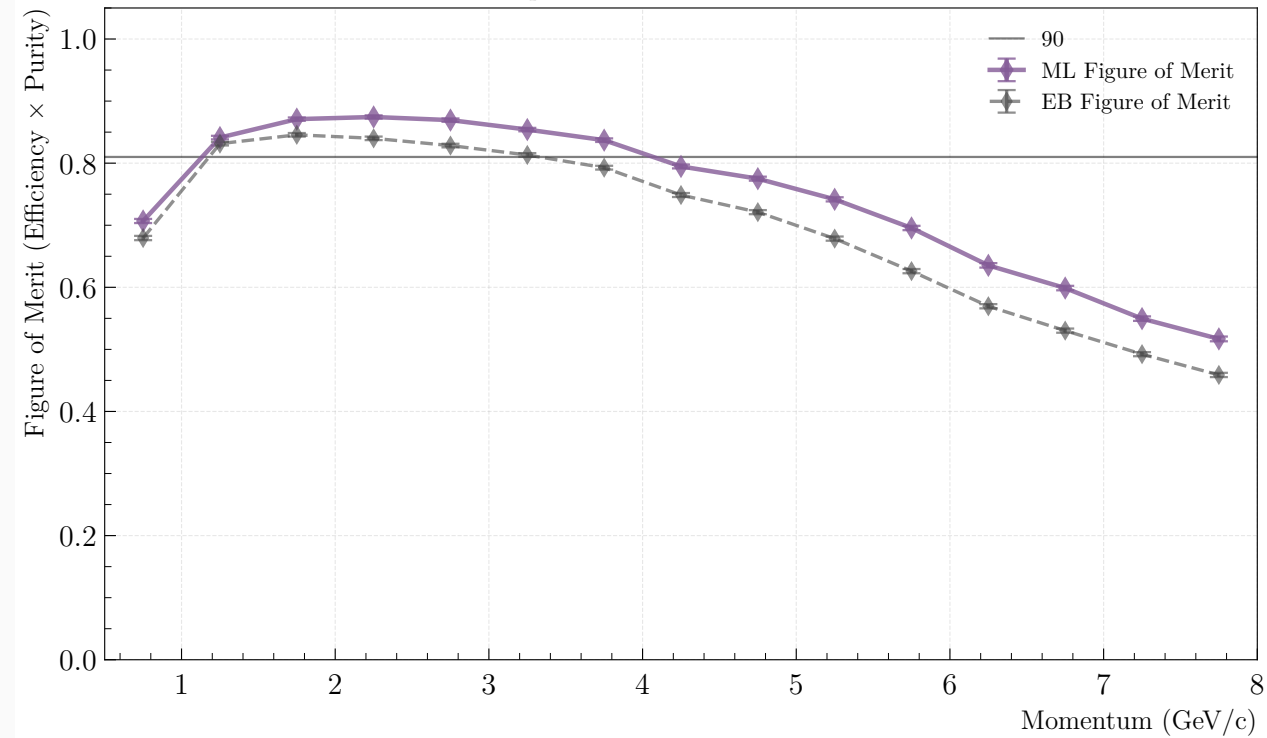
K^- FOM vs Momentum



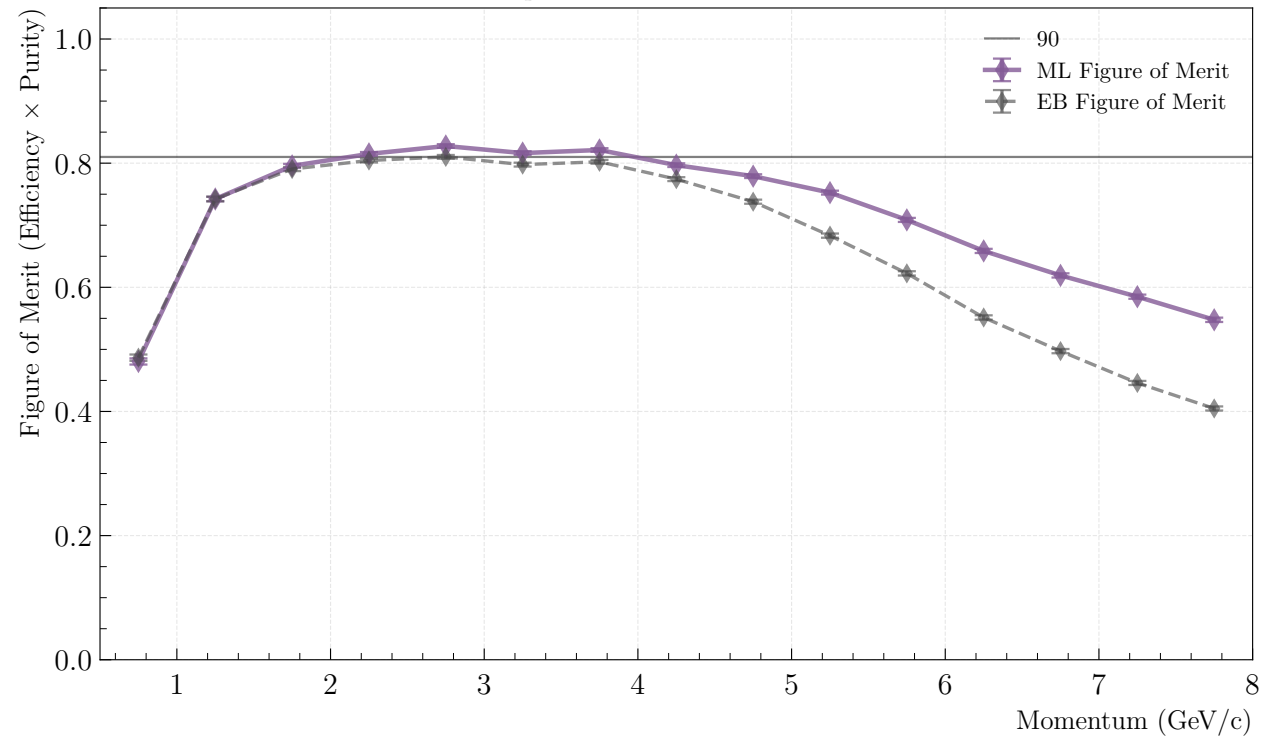
K^+ FOM vs Momentum



$\bar{p}$ FOM vs Momentum



p FOM vs Momentum



Hadron identification: Summary

- With the CatBoost model, we can improve the hadron identification, specially for π and K at high momentum.
- It also improve lower momentum hadron.
- Not much improvement for the proton/anti-proton.

Name	Overall efficiency	Overall purity	Overall FOM
e^+e^-	0.959425	0.964828	0.92568
e^-e^+	0.9641875	0.968846	0.934149
π^-	0.843879	0.873114	0.736803
π^+	0.838079	0.859936	0.720694
K^-	0.78045	0.75665	0.590528
K^+	0.761896	0.75869	0.578043
$\bar{p}$	0.85905	0.848871	0.729222
p	0.851671	0.833633	0.709981